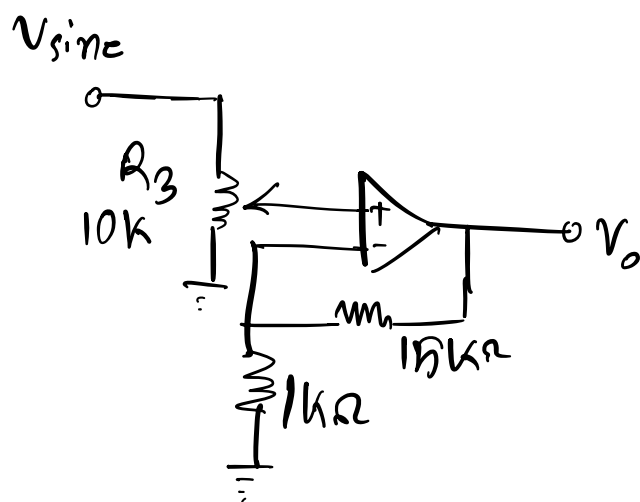


Experiment 1:

For the non-inverting
amp,

$$G = \frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_s}\right)$$
$$= (1 + 15)$$
$$= 16 \text{ V/V}$$



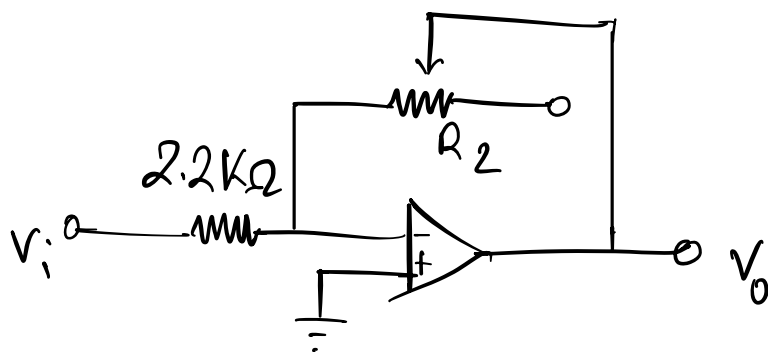
Experiment 3:

For,

$$R_2 = 50 \text{ k}\Omega$$

For inverting op amp,

$$\text{Gain, } \frac{V_o}{V_i} = -\frac{R_f}{R_s} = -\frac{50 \text{ k}}{2.2 \text{ k}} = -22.72 \text{ V/V}$$



For, $R_2 = 59.7 \text{ k}\Omega$ (Potentiometer value where output is at its maximum value without dipping)

$$\text{Gain, } \frac{V_o}{V_i} = -\frac{R_f}{R_s} = -\frac{59.7 \text{ k}}{2.2 \text{ k}} = -27.14 \text{ V/V}$$

Experiment 4:

a) Without the diode,
the $1k\Omega$ path is open,
so no current flows through
there

$$V_a = 15 \cdot \frac{R_2}{R_1 + R_2}$$

For, $R_1 = 2.2k\Omega$ & $R_2 = 6.8k\Omega$,

$$V_a = 15 \cdot \frac{6800}{2200 + 6800}$$

$$= 11.33 \text{ V}$$

For, $R_1 = 4.7k\Omega$ & $R_2 = 4.7k\Omega$,

$$V_a = 7.5 \text{ V}$$

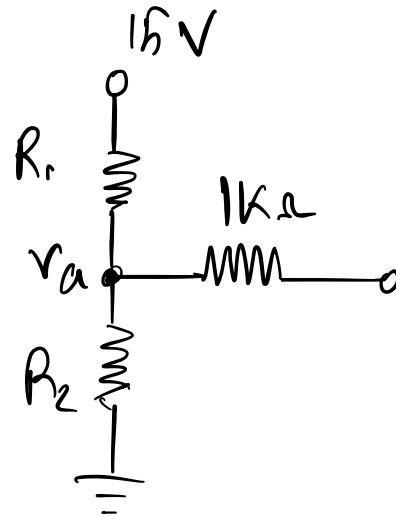
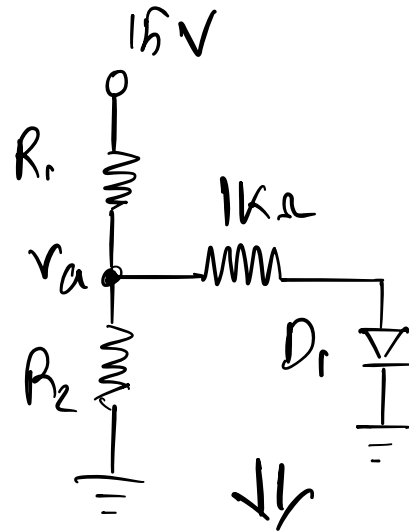
For, $R_1 = 6.8k\Omega$ & $R_2 = 2.2k\Omega$,

$$V_a = 3.67 \text{ V}$$

b) With the diode (LED) $\Rightarrow$

Using KCL,

$$-\frac{15 - V_a}{R_1} + \frac{V_a - 0}{R_2} + \frac{V_a - 2}{1000} = 0$$



$$\Rightarrow \frac{15 - V_a}{R_1} = \frac{V_a}{R_2} + \frac{V_a}{1000} - \frac{2}{1000}$$

$$\Rightarrow 15 - V_a = V_a \left(\frac{R_1}{R_2} + \frac{R_1}{1000} \right) - \frac{2R_1}{1000}$$

$$\Rightarrow 15 + \frac{2R_1}{1000} = \left(\frac{R_1}{R_2} + \frac{R_1}{1000} + 1 \right) V_a$$

$$\Rightarrow V_a = \frac{15 + \frac{2R_1}{1000}}{\frac{R_1}{R_2} + \frac{R_1}{1000} + 1}$$

For, $R_1 = 2.2k\Omega$ & $R_2 = 6.8k\Omega$,

$$V_a = \frac{15 + \frac{2200 \times 2}{1000}}{\left(\frac{2200}{6800} + \frac{2200}{1000} + 1 \right)} = 5.66V$$

For, $R_1 = 4.7k\Omega$ & $R_2 = 4.7k\Omega$,

$$V_a = 4.09V$$

For, $R_1 = 6.8k\Omega$ & $R_2 = 2.2k\Omega$,

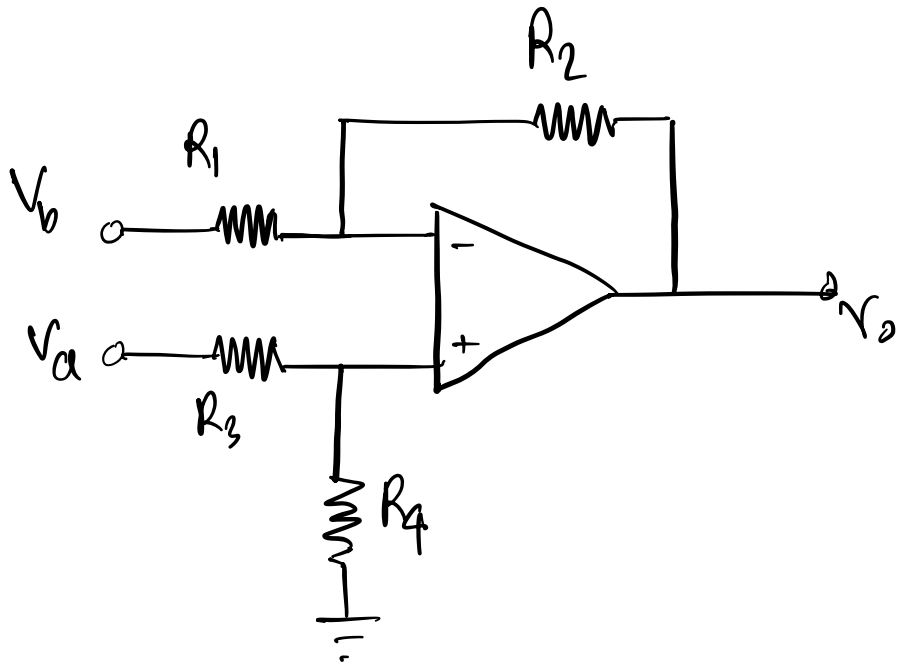
$$V_a = 2.76V$$

Experiment 6:

In the circuit,

$$\frac{R_2}{R_1} = 9.939$$

$$\frac{R_4}{R_3} = 10.009$$



$$V_{DM} = V_a - V_b$$

$$V_{CM} = \frac{V_a - V_b}{2}$$

when $V_b = 0$,

$$V_p = V_a \cdot \frac{R_4}{R_3 + R_4} \quad (\text{voltage divider})$$

$$G = \frac{V_o}{V_a} = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_4}{R_3 + R_4}\right)$$
$$= 9.945$$

When $V_a = 0$,

$$V_o = -\frac{R_2}{R_1} V_b$$

$$G = \frac{V_o}{V_b} = -\frac{R_2}{R_1} = -9.939$$

$$|G| = 9.939$$

$$\text{Common mode gain } (V_a = V_b), G_{CM} = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_4}{R_3 + R_4}\right) - \left(-\frac{R_2}{R_1}\right)$$
$$= 0.006$$

Experiment 1:

$$\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} = \frac{0 - V_o}{R_f}$$

$$V_o = - \left(V_a \cdot \frac{R_f}{R_a} + V_b \cdot \frac{R_f}{R_b} + V_c \cdot \frac{R_f}{R_c} \right)$$

To get -5 from input 101,

$$V_a = 1, V_b = 0, V_c = 1, V_o = 5$$

we get,

$$\frac{R_f}{R_{D2}} + \frac{R_f}{R_{D0}} = 5 \quad \text{--- (i)}$$

Similarly, for 110,

$$\frac{R_f}{R_{D2}} + \frac{R_f}{R_{D1}} = 6 \quad \text{--- (ii)}$$

for 011,

$$\frac{R_f}{R_{D1}} + \frac{R_f}{R_{D0}} = 3 \quad \text{--- (iii)}$$

from (i), (ii) & (iii) $\Rightarrow$

$$\frac{R_f}{R_{D2}} = 4$$

$$\frac{R_f}{R_{D0}} = 1$$

$$\frac{R_f}{R_{D1}} = 2$$

If we take, $R_f = 4k\Omega$

then we need, $R_{D2} = 1k\Omega$, $R_{D1} = 2k\Omega$, $R_{D0} = 4k\Omega$